

Digital Communications

Spread-Spectrum Techniques

- Pseudo-Noise Sequences
- Maximal-length sequences
- Understanding Spread Spectrum
- Direct-Sequence Spread Spectrum (DSSS)
- Frequency Hopping Spread Spectrum (FHSS)
- Code Division Multiplexing – CDM
- Code Division Multiple Access - CDMA



Introduction

- The techniques considered in this session are called spread spectrum because the transmission bandwidth employed is much greater than the minimum bandwidth required to transmit the information
- A system is defined to be a spread-spectrum system if it fulfills the following requirements:
 1. The signal occupies a bandwidth much in excess of the minimum bandwidth necessary to send the information

Introduction

2. Spreading is accomplished by means of a spreading signal, often called a code signal, which is independent of the data
3. At the receiver, despreading (recovering the original data) is accomplished by the correlation of the received spread signal with a synchronized replica of the spreading signal used to spread the information
 - Standard modulation schemes such as frequency modulation and pulse code modulation also spread the spectrum of an information signal, but they do not qualify as spread-spectrum systems since they do not satisfy all the conditions outlined above



Introduction

- In spread-spectrum modulation, channel bandwidth and transmit power are sacrificed for the sake of secure communications
- Specifically, we cover the following topics:
 - Spreading sequences in the form of pseudo-noise sequences, their properties, and methods of generation
 - The basic notion of spread-spectrum modulation
 - The two commonly used types of spread-spectrum modulation: direct sequence and frequency hopping



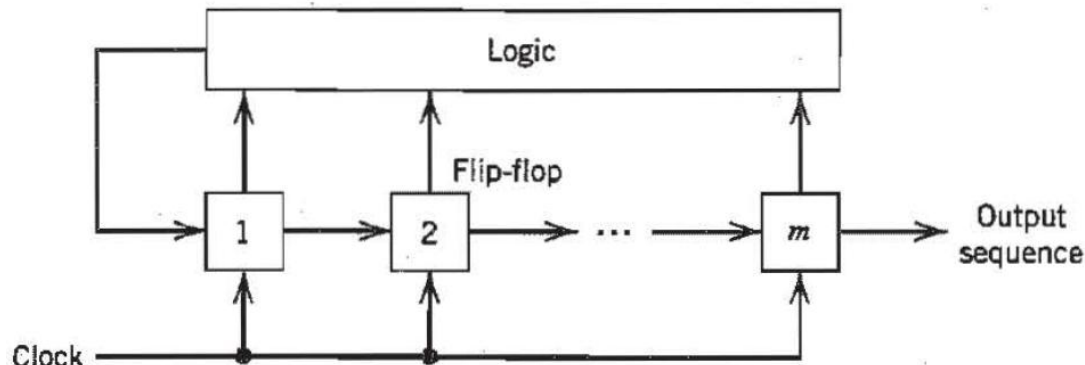
Introduction

- For their operation, both of these techniques rely on the availability of a noise-like spreading code called a pseudo-random or pseudo-noise sequence
- Since such a sequence is basic to the operation of spread-spectrum modulation, it is logical that we begin our study by describing the generation and properties of pseudo-noise sequences



Pseudo-Noise Sequences

- A pseudo-noise (PN) sequence is a periodic binary sequence with a noise-like waveform that is usually generated by means of a feedback shift register



- A feedback shift register consists of an ordinary shift-register made up of m flip-flops (two-state memory stages) and a logic circuit that are interconnected to form a multiloop feedback circuit

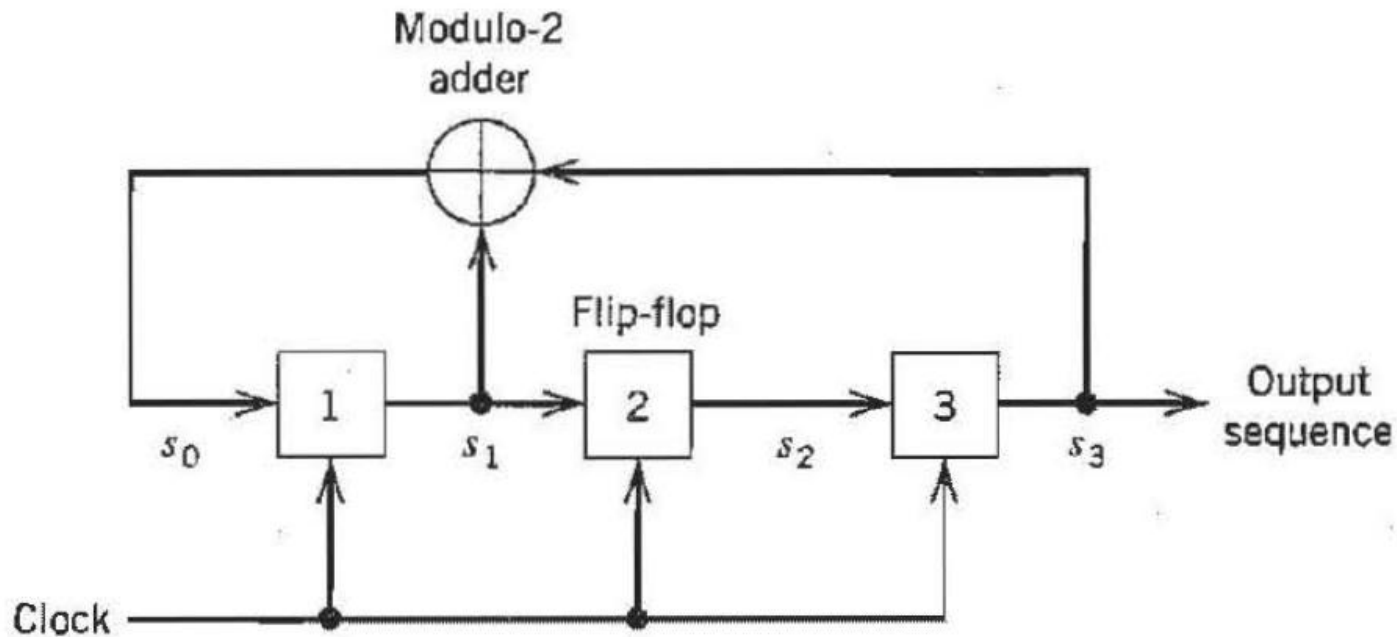
Pseudo-Noise Sequences

- The flip-flops in the shift register are regulated by a single timing clock
- At each pulse (tick) of the clock, the state of each flip-flop is shifted to the next one down the line
- With each clock pulse the logic circuit computes a boolean function of the states of the flip-flops
- The result is then fed back as the input to the first flip-flop, thereby preventing the shift register from emptying
- The PN sequence so generated is determined by the length m of the shift register, its initial state, and the feedback logic



Pseudo-Noise Sequences

- Example:
- Consider the linear feedback shift register involving three flip-flops:



Pseudo-Noise Sequences

- Example:
- The input s_0 applied to the first flip-flop is equal to the modulo-2 sum of s_1 and s_3
- It is assumed that the initial state of the shift register is 100 (reading the contents of the three flip-flops from left to right). Then, the succession of states will be as follows:
 - 100, 110, 111, 011, 101, 010, 001, 100, ...



Pseudo-Noise Sequences

- Example:
- The output sequence (the last position of each state of the shift register) is therefore:
 - 0 0 1 1 1 0 1 0 ...
- which repeats itself with period $2^3 - 1 = 7$
 - Note that the choice of 100 as the initial state is arbitrary
 - Any of the other six permissible states could serve equally well as an initial state, except 000 state



Pseudo-Noise Sequences

- A feedback shift register is said to be linear when the feedback logic consists entirely of modulo-2 adders
 - In such a case, the zero state (e.g., the state for which all the flip-flops are in state 0) is not permitted
 - For a zero state, the input s_0 produced by the feedback logic would be 0, the shift register would then continue to remain in the zero state, and the output would therefore consist entirely of 0s

Maximal-length sequences

- Consequently, the period of a PN sequence produced by a linear feedback shift register with m flip-flops cannot exceed $2^m - 1$
- When the period is exactly $2^m - 1$, the PN sequence is called a maximal-length sequence or simply m -sequence
- Maximal-length sequences have many of the properties possessed by a truly random binary sequence



Maximal-length sequences

- A random binary sequence is a sequence in which the presence of binary symbol 1 or 0 is equally probable
- Some properties of maximal-length sequences are as follows:
 1. In each period of a maximal-length sequence, the number of 1s is always one more than the number of 0s. This property is called the balance property



Maximal-length sequences

2. Among the runs of 1s and of 0s in each period of a maximal-length sequence, one half the runs, of each kind are of length one, one-fourth are of length two, and so on, as long as these fractions represent meaningful numbers of runs
 - This property is called the run property. By a "run" we mean a subsequence of identical symbols (1s or 0s) within one period of the sequence
 - The length of this subsequence is the length of the run
 - For a maximal-length sequence generated by a linear feedback shift register of length m , the total number of runs is $(N + 1)/2$, where $N = 2^m - 1$



Maximal-length sequences

3. The autocorrelation function of a maximal-length sequence is periodic and binary valued
 - This property is called the correlation property
 - The period of a maximum-length sequence is defined by:

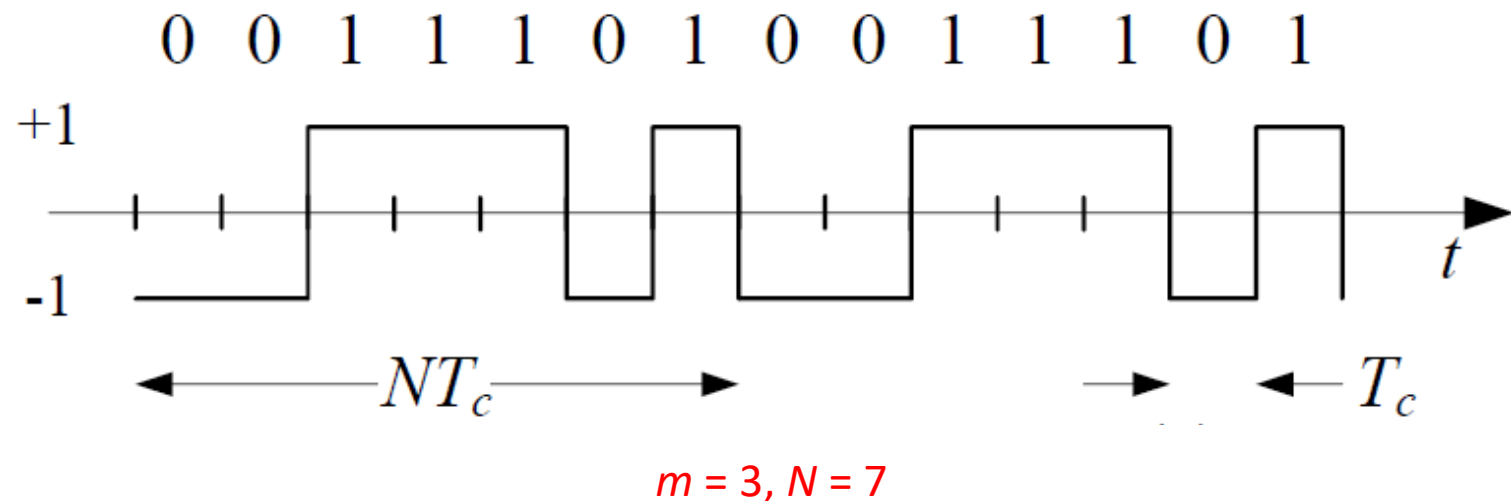
$$N = 2^m - 1$$

- where m is the length of the shift register



Maximal-length sequences

- Let binary symbols 0 and 1 of the sequence be denoted by the levels -1 and +1, respectively
- Let $c(t)$ denote the resulting waveform of the maximal-length sequence, as illustrated above:



Maximal-length sequences

- The period of the waveform $c(t)$ is:

$$T_b = NT_c$$

- where T_c is the duration assigned to symbol 1 or 0 in the maximal-length sequence
- By definition, the autocorrelation function of a periodic signal $c(t)$ of period T_b is:

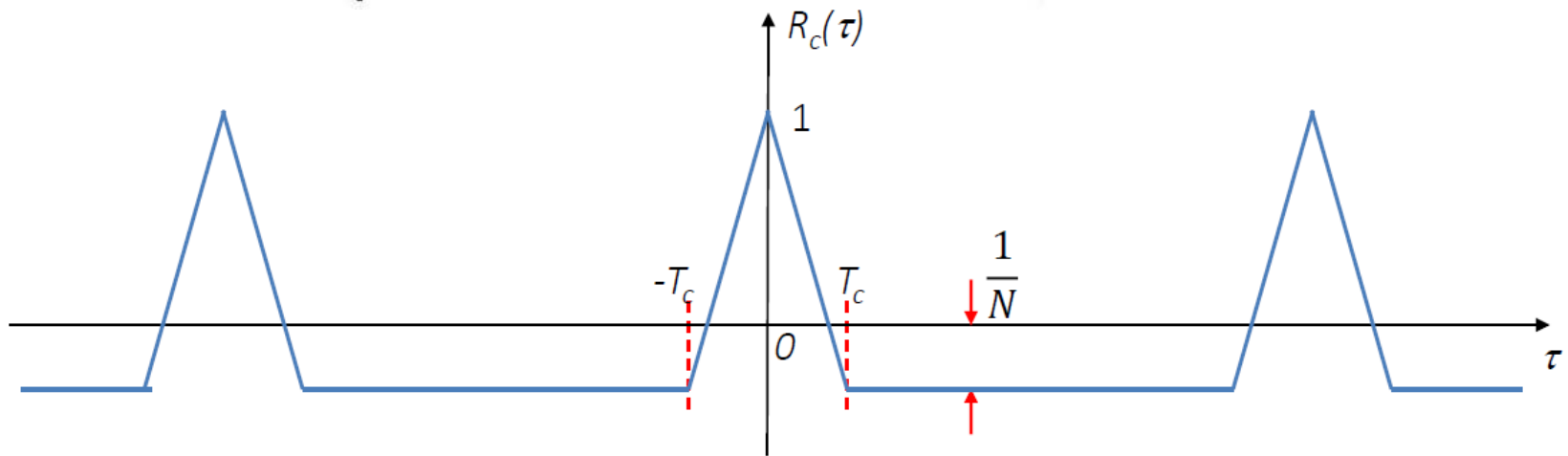
$$R_c(\tau) = \frac{1}{T_b} \int_{-T_b/2}^{T_b/2} c(t)c(t - \tau) dt$$



Maximal-length sequences

- Applying this formula to a maximal-length sequence represented by $c(t)$, we get:

$$R_c(\tau) = \begin{cases} 1 - \frac{N+1}{NT_c} |\tau|, & |\tau| \leq T_c \\ -\frac{1}{N}, & \text{for the remainder of the period} \end{cases}$$



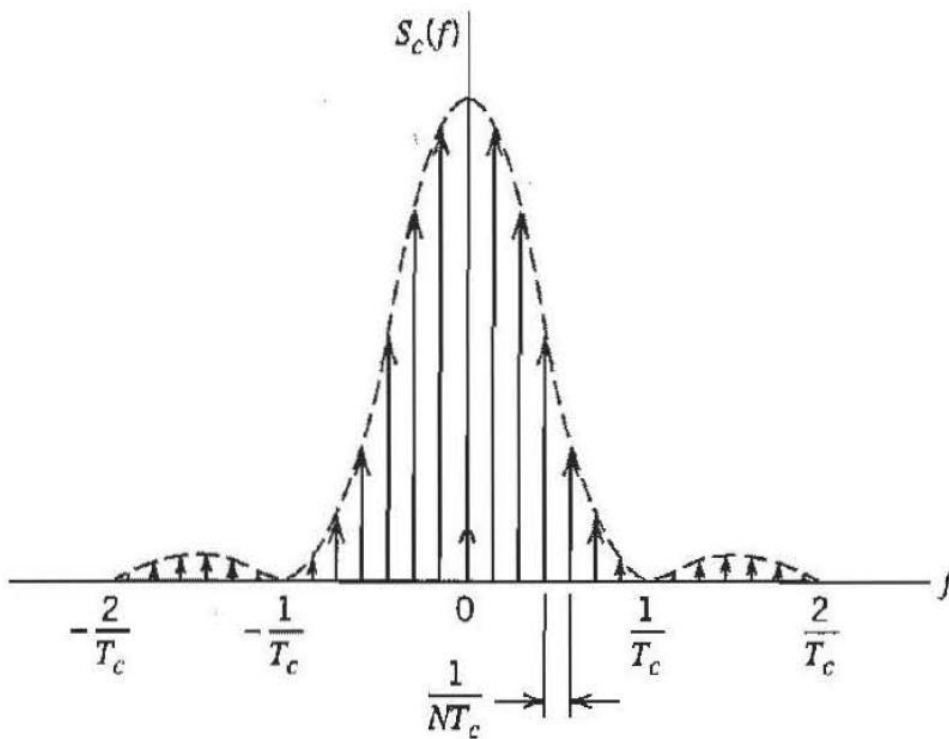
Maximal-length sequences

- From Fourier transform theory we know that periodicity in the time domain is transformed into uniform sampling in the frequency domain
- This interplay between the time and frequency domains is borne out by the power spectral density of the maximal-length wave $c(t)$
- Specifically, taking the Fourier transform of previous equation, we get the sampled spectrum:

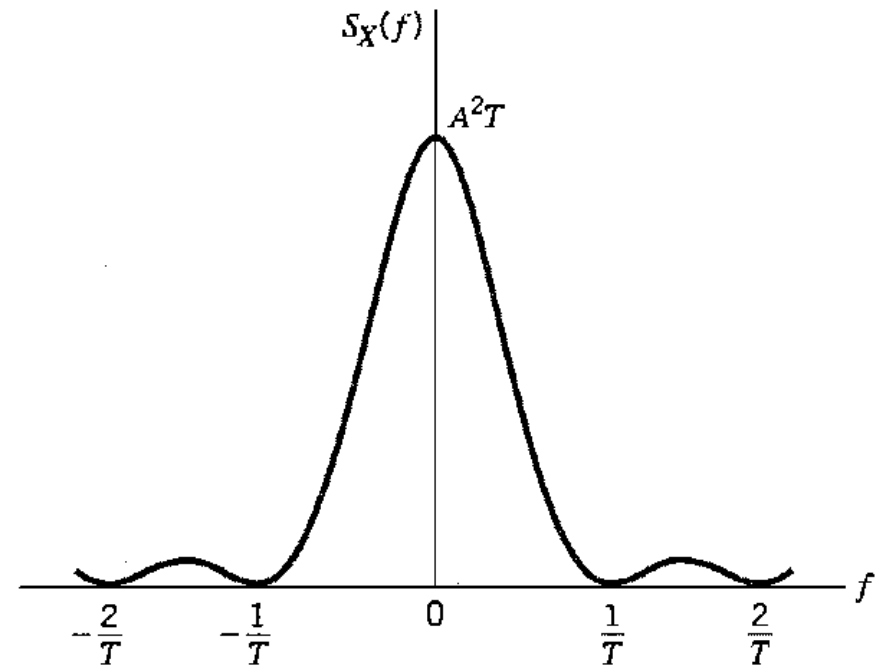
$$S_c(f) = \frac{1}{N^2} \delta(f) + \frac{1+N}{N^2} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \text{sinc}^2\left(\frac{n}{N}\right) \delta\left(f - \frac{n}{NT_c}\right)$$



Maximal-length sequences

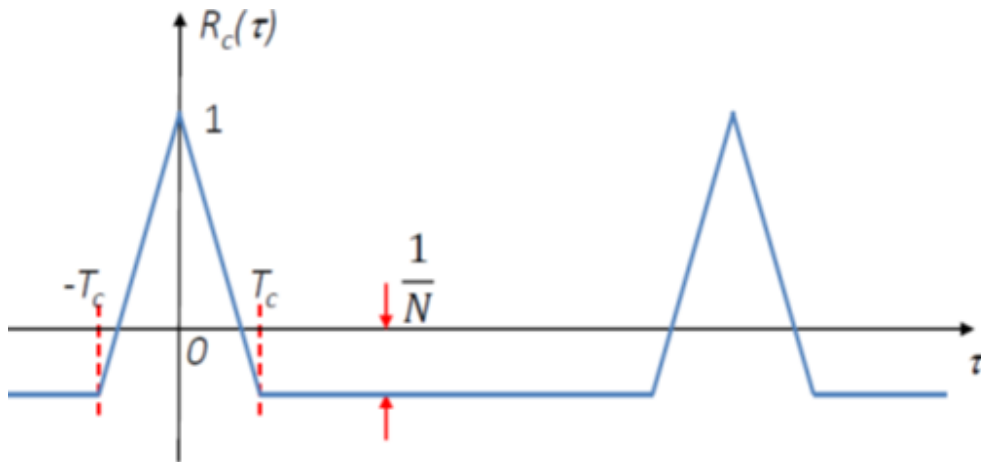


Power spectral density of
maximal-length sequence

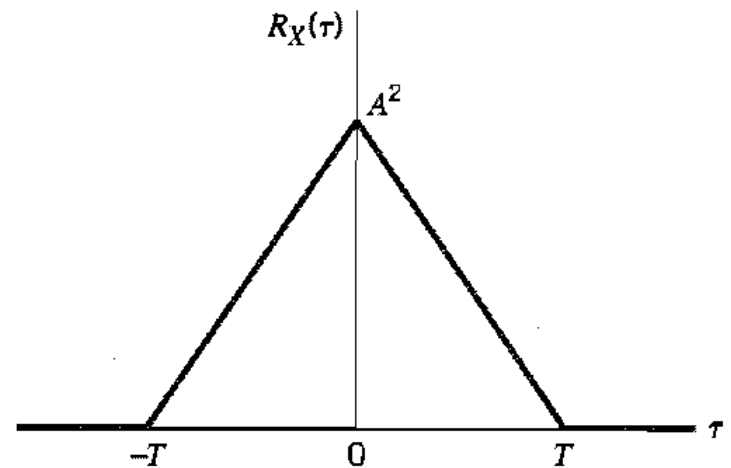


Power spectral density of
random binary wave

Maximal-length sequences



Autocorrelation function of maximal-length sequence



Autocorrelation function of random binary wave

Maximal-length sequences

- As the shift-register length m , or equivalently, the period N of the maximal-length sequence is increased, the maximal-length sequence becomes increasingly similar to the random binary sequence
- Indeed, in the limit, the two sequences become identical when N is made infinitely large
- However, the price paid for making N large is an increasing storage requirement, which imposes a practical limit on how large N can actually be made



Maximal-length sequences

- Comparing the results for a maximal-length sequence with the corresponding results shown for a random binary sequence, we may make the following observations:
 - For a period of the maximal-length sequence, the autocorrelation function $R_c(\tau)$ is somewhat similar to that of a random binary wave
 - The waveforms of both sequences have the same envelope, $\text{sinc}^2(fT)$, for their power spectral densities
 - The fundamental difference is that whereas the random binary sequence has a continuous spectral density characteristic, the corresponding characteristic of a maximal-length sequence consists of *dirac* functions spaced $1/NT_c$ Hz apart



Maximal-length sequences

- Choosing a maximal-length sequence
- The key question that we need to address is:
 - How do we find the feedback logic for a desired period N ?
 - The answer to this question is to be found in the theory of error-control codes
 - The task of finding the required feedback logic is made particularly easy for us by virtue of the extensive tables of the necessary feedback connections for varying shift-register lengths that have been compiled in the literature



Maximal-length sequences

- Choosing a maximal-length sequence

<i>Shift-Register Length, m</i>	<i>Feedback Taps</i>
2*	[2, 1]
3*	[3, 1]
4	[4, 1]
5*	[5, 2], [5, 4, 3, 2], [5, 4, 2, 1]
6	[6, 1], [6, 5, 2, 1], [6, 5, 3, 2]
7*	[7, 1], [7, 3], [7, 3, 2, 1], [7, 4, 3, 2], [7, 6, 4, 2], [7, 6, 3, 1], [7, 6, 5, 2], [7, 6, 5, 4, 2, 1], [7, 5, 4, 3, 2, 1]
8	[8, 4, 3, 2], [8, 6, 5, 3], [8, 6, 5, 2], [8, 5, 3, 1], [8, 6, 5, 1], [8, 7, 6, 1], [8, 7, 6, 5, 2, 1], [8, 6, 4, 3, 2, 1]

- Note that as m increases, the number of alternative schemes (codes) is enlarged



Understanding Spread Spectrum

- Spread-spectrum modulation can provide protection against externally generated interfering (jamming) signals with finite power
- The jamming signal may consist of a fairly powerful broadband noise or multitone waveform that is directed at the receiver for the purpose of disrupting communications
- Protection against jamming waveforms is provided by making the information-bearing signal occupy a bandwidth far in excess of the minimum bandwidth necessary to transmit it
- The resulting effect causes the transmitted signal to take on a noise-like appearance so as to blend in with the background



Understanding Spread Spectrum

- The transmitted signal is thus enabled to propagate through the channel undetected by anyone who may be listening
- We may therefore think of spread spectrum as a method of "camouflaging" the information-bearing signal
- One method of widening the bandwidth of an information-bearing (data) sequence involves the use of modulation

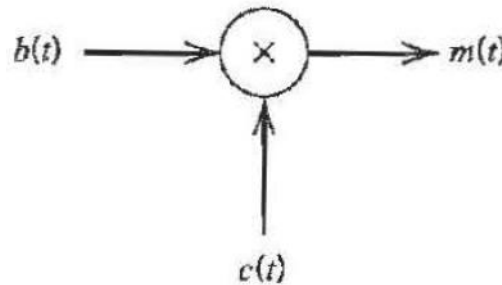


Understanding Spread Spectrum

- Let $\{b_k\}$ denote a binary data sequence, and $\{c_k\}$ denote a pseudo-noise (PN) sequence
- Let the waveforms $b(t)$ and $c(t)$ denote their respective polar NRZ representations in terms of two levels equal in amplitude and opposite in polarity, namely, ± 1
- We will refer to $b(t)$ as the information-bearing (data) signal, and to $c(t)$ as the PN signal
- The desired modulation is achieved by applying the data signal $b(t)$ and the PN signal $c(t)$ to a product modulator or multiplier

Understanding Spread Spectrum

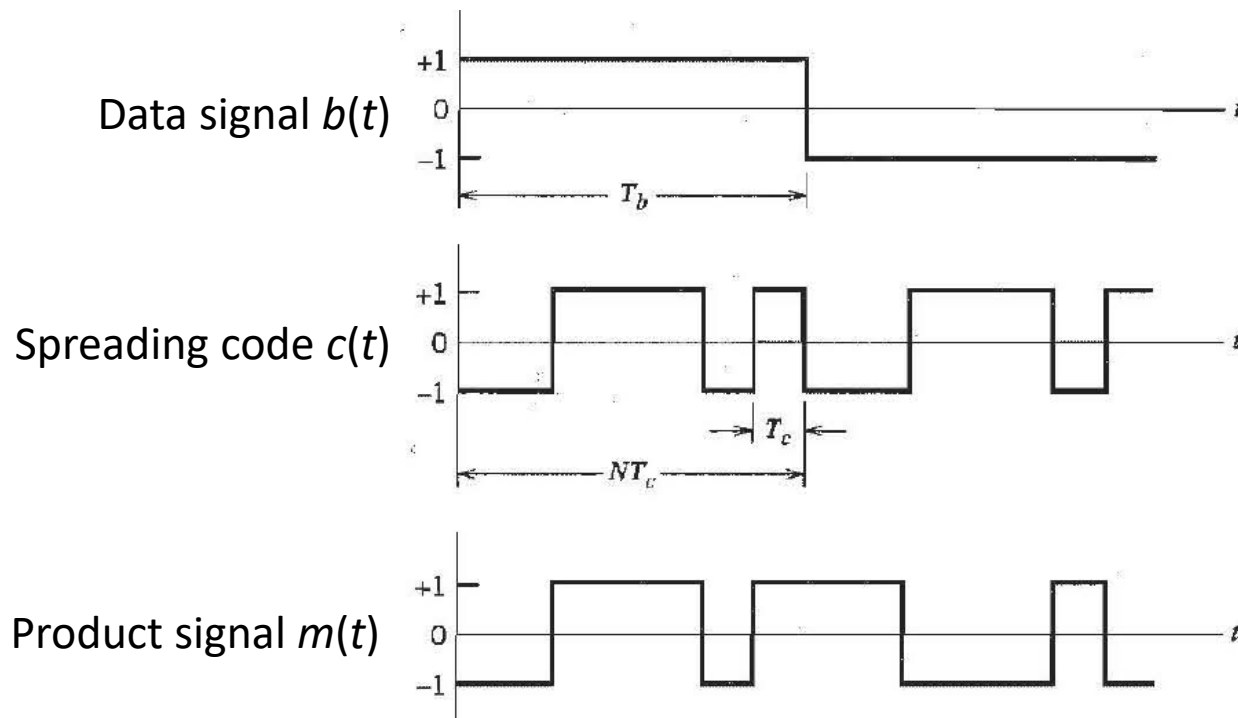
- Idealized model of baseband spread-spectrum system – Tx



- From Fourier transform theory, multiplication of two signals produces a signal whose spectrum equals the convolution of the spectra of the two component signals
- If the message signal $b(t)$ is narrowband and the PN signal $c(t)$ is wideband, the product signal $m(t)$ will have a spectrum that is nearly the same as the wideband PN signal

Understanding Spread Spectrum

- By multiplying the information-bearing signal $b(t)$ by the PN signal $c(t)$, each information bit is "chopped" up into a number of small time increments



Understanding Spread Spectrum

- These small time increments are commonly referred to as “chips”
- For baseband transmission, the product signal $m(t)$ represents the transmitted signal
- We may thus express the transmitted signal as:

$$m(t) = c(t)b(t)$$

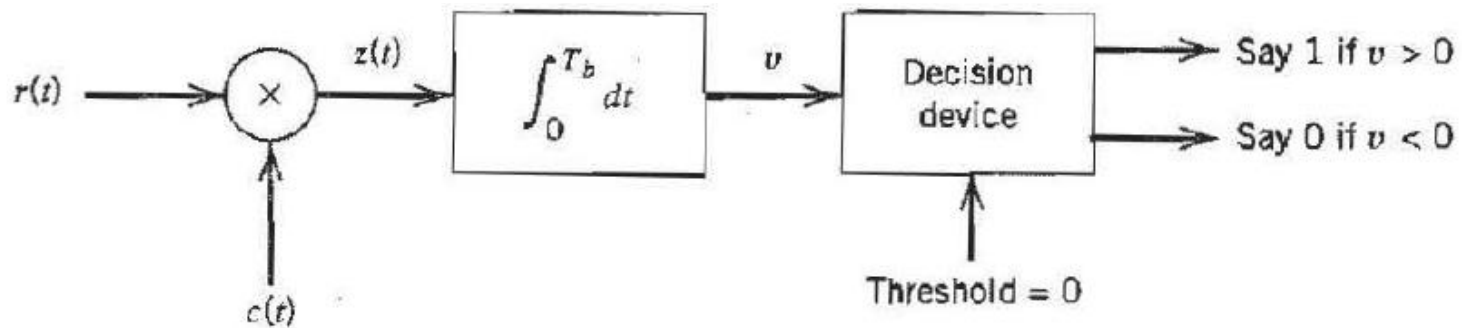
- The received signal $r(t)$ consists of the transmitted signal $m(t)$ plus an additive interference denoted by $i(t)$:

$$r(t) = m(t) + i(t) = c(t)b(t) + i(t)$$



Understanding Spread Spectrum

- To recover the original message signal $b(t)$, the received signal $r(t)$ is applied to a demodulator that consists of a multiplier followed by an integrator, and a decision device



- The multiplier is supplied with a locally generated PN sequence that is an exact replica of that used in the transmitter

Understanding Spread Spectrum

- Moreover, we assume that the receiver operates in perfect synchronism with the transmitter, which means that the PN sequence in the receiver is lined up exactly with that in the transmitter
- The multiplier output in the receiver is therefore given by:

$$z(t) = c(t)r(t) = c^2(t)b(t) + c(t)i(t)$$

- Data signal $b(t)$ is multiplied twice by the PN signal $c(t)$, whereas the unwanted signal $i(t)$ is multiplied only once



Understanding Spread Spectrum

- The PN signal $c(t)$ alternates between the levels -1 and +1, and the alternation is destroyed when it is squared, hence:

$$c^2(t) = 1$$

- And:

$$z(t) = b(t) + c(t)i(t)$$

- We thus see that, the data signal $b(t)$ is reproduced at the multiplier output in the receiver, except for the effect of the interference represented by the additive term $c(t).i(t)$



Understanding Spread Spectrum

- Multiplication of the interference $i(t)$ by the locally generated PN signal $c(t)$ means that the spreading code will affect the interference just as it did the original signal at the transmitter
- The data component $b(t)$ is narrowband, whereas the spurious component $c(t).i(t)$ is wideband
- Applying the multiplier output to a baseband LPF with a bandwidth just large enough to accommodate the recovery of the data signal $b(t)$, most of the power in the spurious component $c(t).i(t)$ is filtered out



Understanding Spread Spectrum

- The use of a spreading code (with pseudo-random properties) in the transmitter produces a wideband transmitted signal that appears noiselike to a receiver that has no knowledge of the spreading code
- The longer we make the period of the spreading code, the closer will the transmitted signal be to a truly random binary wave, and the harder is to detect
- The price we have to pay for the improved protection against interference is increased transmission bandwidth, system complexity, and processing delay



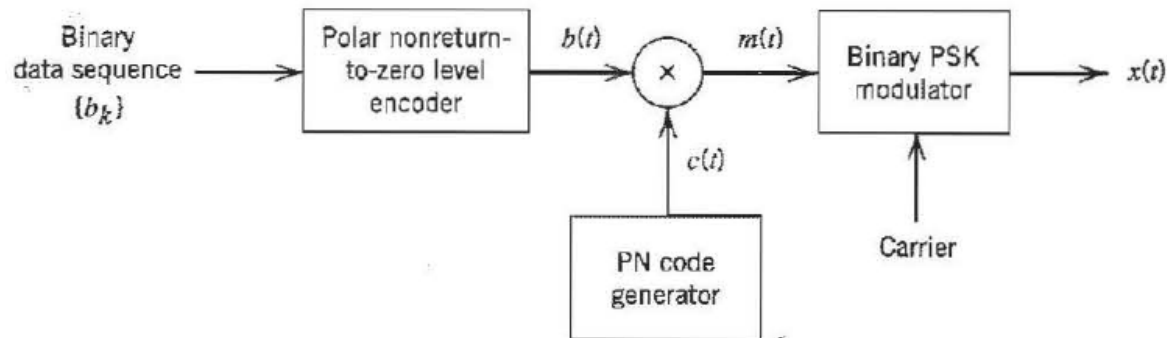
Direct Sequence Spread Spectrum

- The spread-spectrum technique described until now is referred to as Direct Sequence Spread Spectrum – DSSS
- The discussion presented was in the context of baseband transmission
- To provide for the use of this technique in passband transmission over a satellite channel, for example, we may incorporate Coherent Binary Phase-Shift Keying (PSK) into the transmitter and receiver



Direct Sequence Spread Spectrum

- The transmitter first converts the incoming binary data sequence $\{b_k\}$ into a polar NRZ waveform $b(t)$, which is followed by two stages of modulation



- The first stage consists of a product modulator with the data signal $b(t)$ (representing a data sequence) and the PN signal $c(t)$ (representing the PN sequence) as inputs
- The second stage consists of a binary PSK modulator

Direct Sequence Spread Spectrum

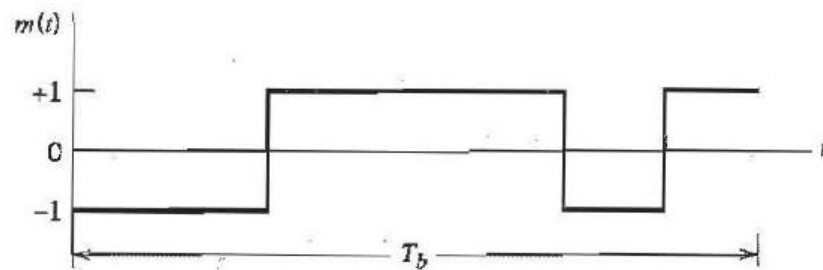
- The transmitted signal $x(t)$ is thus a direct-sequence spread binary phase-shift-keyed (DS/BPSK) signal
- The phase modulation $\theta(t)$ of $x(t)$ has one of two values, 0 and π , depending on the polarities of the message signal $b(t)$ and PN signal $c(t)$ at time t in accordance with the truth table (view slide 30):

		<i>Polarity of Data Sequence $b(t)$ at Time t</i>	
		$+$	$-$
<i>Polarity of PN sequence $c(t)$ at time t</i>	$+$	0	π
	$-$	π	0

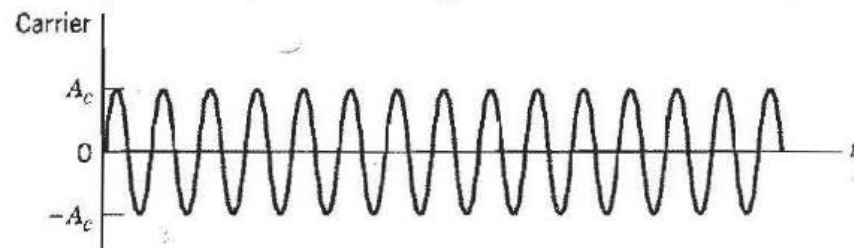
Direct Sequence Spread Spectrum

- Next figure illustrates the waveforms for the second stage of modulation to one period of the PN sequence:

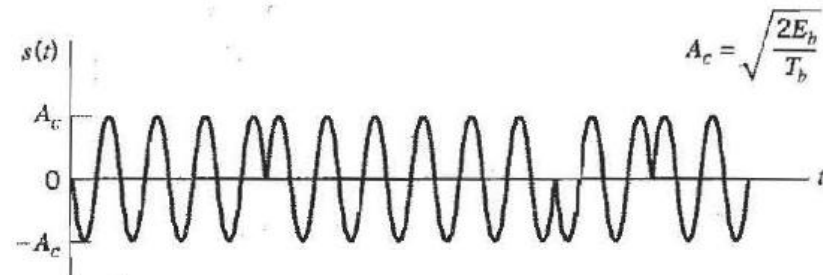
Product signal $m(t) = c(t).b(t)$



Sinusoidal carrier

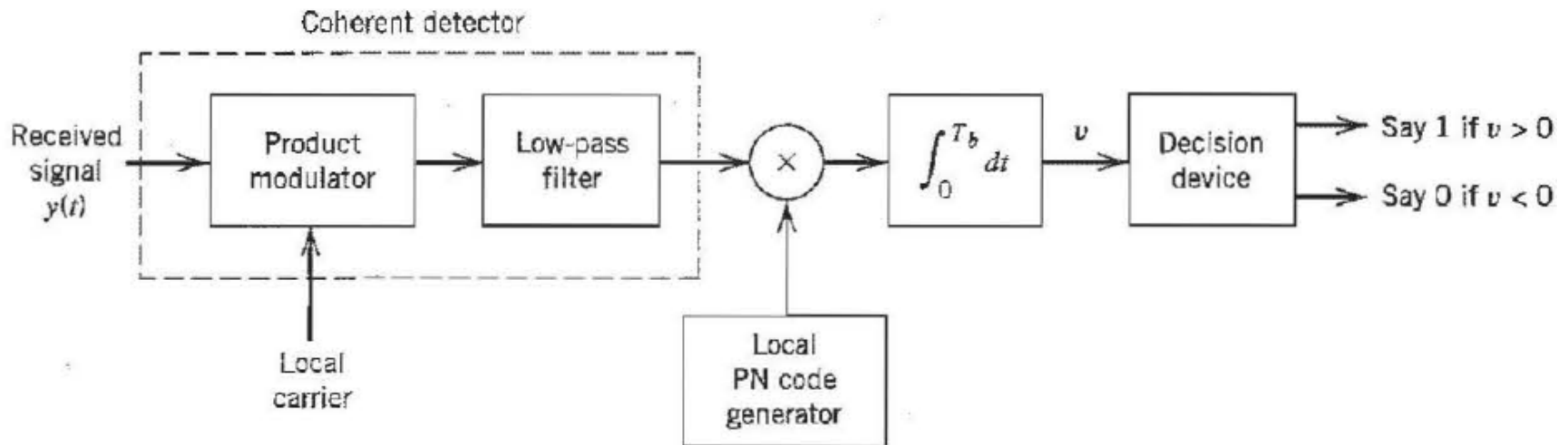


DS/BPSK signal



Direct Sequence Spread Spectrum

- The receiver consists of two stages of demodulation
- In the first stage, the received signal $y(t)$ and a locally generated carrier are applied to a product modulator followed by a LPF whose bandwidth is equal to that of the original message signal $m(t)$



Direct Sequence Spread Spectrum

- This stage reverses the phase-shift keying applied to the transmitted signal
- The second stage of demodulation performs spectrum despreading by multiplying the LPF output by a locally generated replica of the PN signal $c(t)$, followed by integration over a bit interval $0 < t < T_b$
- Finally decision is making in the manner described in slides 32 - 36



Frequency Hopping Spread Spectrum

- Until now, the use of a PN sequence to modulate a phase-shift-keyed signal achieves instantaneous spreading of the transmission bandwidth
- The ability of such a system to combat the effects of jammers is determined by the processing gain of the system, which is a function of the PN sequence period
- The processing gain can be made larger by employing a PN sequence with narrow chip duration, which, in turn, permits a greater transmission bandwidth and more chips per bit

Frequency Hopping Spread Spectrum

- However, the capabilities of physical devices used to generate the PN spread-spectrum signals impose a practical limit on the attainable processing gain
- Indeed, it may turn out that the processing gain so attained is still not large enough to overcome the effects of some jammers of concern, in which case we have to resort to other methods
- One such alternative method is to force the jammer to cover a wider spectrum by randomly hopping the data-modulated carrier from one frequency to the next



Frequency Hopping Spread Spectrum

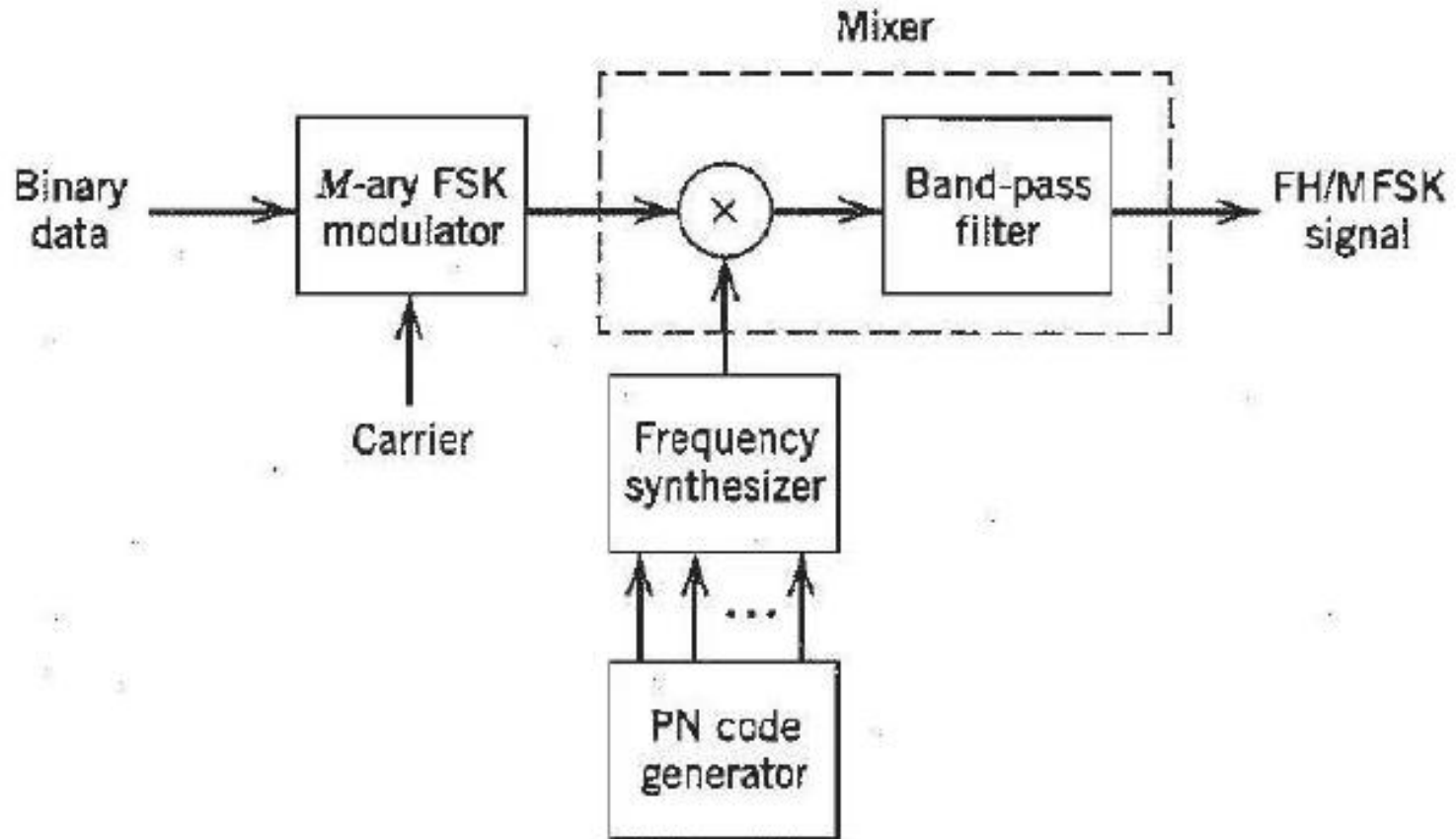
- In effect, the spectrum of the transmitted signal is spread sequentially rather than instantaneously; the term "sequentially" refers to the pseudo-random-ordered sequence of frequency hops
- The type of spread spectrum in which the carrier hops randomly from one frequency to another is called Frequency-Hopping (FH) spread spectrum
- A common modulation format for FH systems is that of *M*-ary Frequency-Shift Keying (MFSK). The combination of these two techniques is referred to simply as FH/MFSK

Frequency Hopping Spread Spectrum

- Since FH does not cover the entire spread spectrum instantaneously, we are led to consider the rate at which the hops occur
- In this context, we may identify two basic (technology-independent) characterizations of FH:
 - 1. Slow-FH, in which the symbol rate R_s of the MFSK signal is an integer multiple of the hop rate R_h . That is, several symbols are transmitted on each frequency hop
 - 2. Fast-FH, in which the hop rate R_h is an integer multiple of the MFSK symbol rate R_s . That is, the carrier frequency will change or hop several times during the transmission of one symbol

Slow Frequency Hopping

- FH spread M -ary frequency-shift keying - Transmitter



Slow Frequency Hopping

- A FH/MFSK transmitter, involves frequency modulation followed by mixing
- First, the incoming binary data are applied to an M -ary FSK modulator
- The resulting modulated wave and the output from a digital frequency synthesizer are then applied to a mixer that consists of a multiplier followed by a band-pass filter
- The filter is designed to select the sum frequency component resulting from the multiplication process as the transmitted signal

Slow Frequency Hopping

- In particular, successive k -bit segments of a PN sequence drive the frequency synthesizer, which enables the carrier frequency to hop over 2^k distinct values
- On a single hop, the bandwidth of the transmitted signal is the same as that resulting from the use of a conventional MFSK with an alphabet of $M = 2^k$ orthogonal signals
- However, for a complete range of 2^k frequency hops, the transmitted FH/MFSK signal occupies a much larger bandwidth



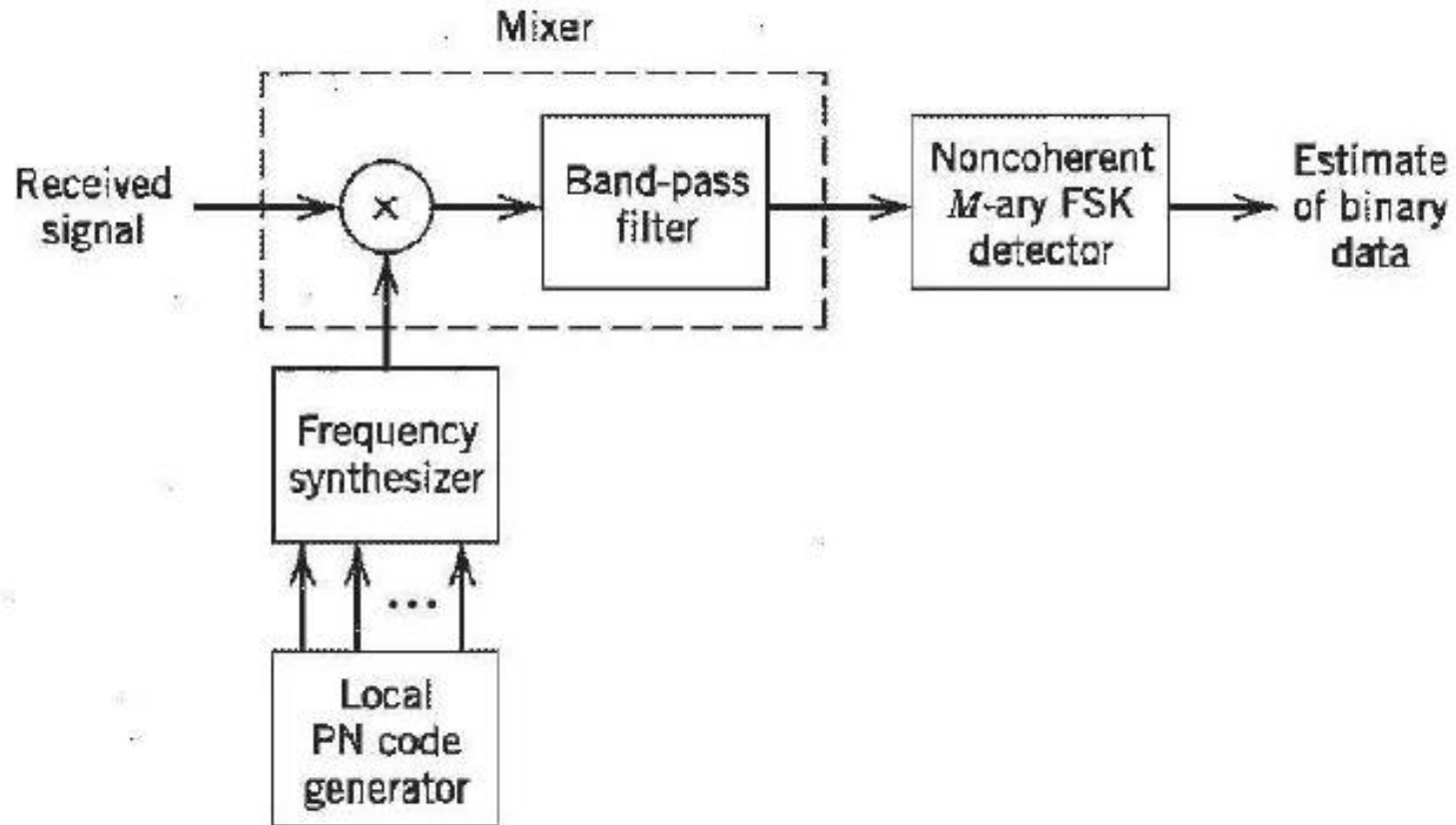
Slow Frequency Hopping

- Indeed, with present-day technology, FH bandwidths on the order of several GHz are attainable, which is an order of magnitude larger than that achievable with DSSS
- An implication of these large FH bandwidths is that coherent detection is possible only within each hop, because frequency synthesizers are unable to maintain phase coherence over successive hops
- Accordingly, most FHSS communication systems use noncoherent M -ary modulation schemes



Slow Frequency Hopping

- FH spread M -ary frequency-shift keying - Receiver



Slow Frequency Hopping

- In the receiver the FH is first removed by mixing (down-converting) the received signal with the output of a local frequency synthesizer that is synchronously controlled in the same manner as that in the transmitter
- The resulting output is then band-pass filtered, and subsequently processed by a noncoherent M -ary FSK detector
- To implement this M -ary detector, we may use a bank of M noncoherent matched filters, each of which is matched to one of the MFSK tones
- An estimate of the original symbol transmitted is obtained by selecting the largest filter output



Slow Frequency Hopping

- An individual FH/MFSK tone of shortest duration is referred to as a “chip”
 - this terminology should not be confused with that used when describing DSSS/BPSK
- The chip rate, R_c for an FH/MFSK system is defined by:

$$R_c = \max(R_h, R_s)$$

- where R_h is the hop rate, and R_s is the symbol rate
- A slow FH/MFSK signal is characterized by having multiple symbols transmitted per hop. Hence, each symbol of a slow FH/MFSK signal is a chip

Slow Frequency Hopping

- Correspondingly, in a slow FH/MFSK system, the bit rate R_b of the incoming binary data, the symbol rate R_s of the MFSK signal, the chip rate R_c and the hop rate R_h are related by:

$$R_c = R_s = \frac{R_b}{K} \geq R_h$$

– where $K = \log_2 M$

- At each hop, the MFSK tones are separated in frequency by an integer multiple of the chip rate $R_c = R_s$, ensuring their orthogonality



Slow Frequency Hopping

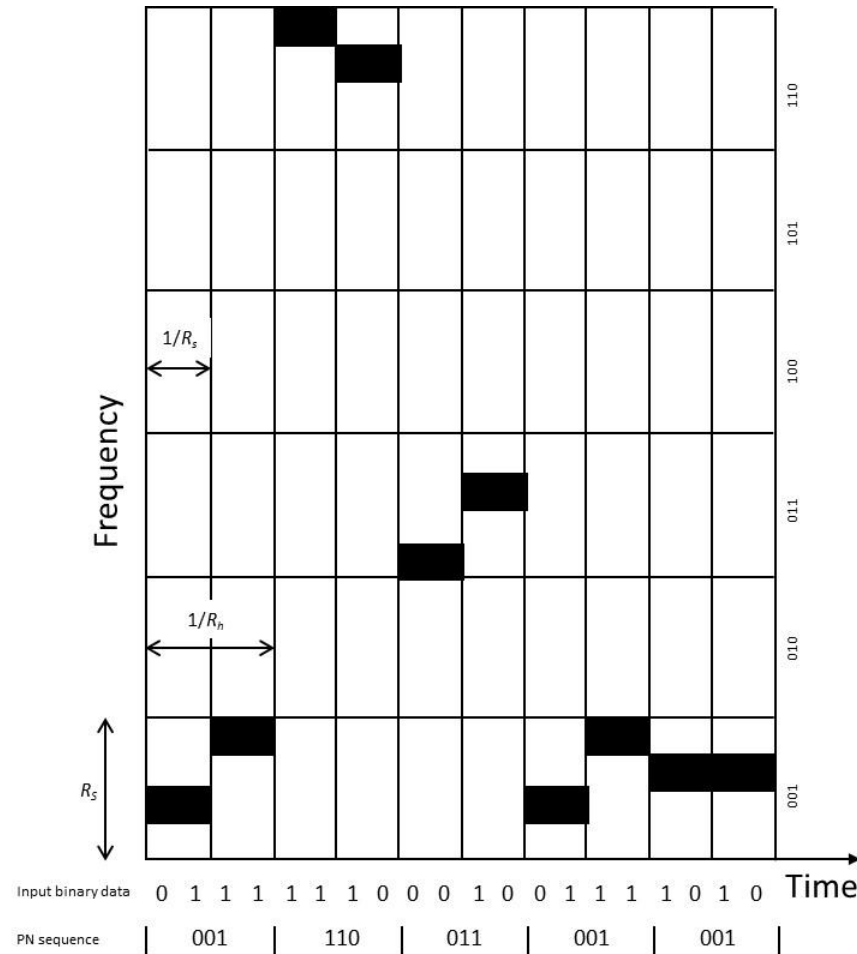
- Example:
- Variation of the frequency of a slow FH/MFSK signal with time for one complete period of the PN sequence
- The period of the PN sequence is $2^4 - 1 = 15$
- The FH/MFSK signal has the following parameters:

Number of bits per MFSK symbol	$K = 2$
Number of MFSK tones	$M = 2^K = 4$
Length of PN segment per hop	$k = 3$
Total number of frequency hops	$2^k = 8$



Slow Frequency Hopping

- Example:



Slow Frequency Hopping

- Example:
- In this example, the carrier is hopped to a new frequency after transmitting two symbols or equivalently, four information bits
- Figure also includes the input binary data, and the PN sequence controlling the selection of FH carrier frequency
- It is noteworthy that although there are eight distinct frequencies available for hopping, only three of them are utilized by the PN sequence

Slow Frequency Hopping

- Next figure shows the variation of the dehopped frequency with time:



- This variation is recognized to be the same as that of a conventional MFSK signal produced by the given input data

Fast Frequency Hopping

- A fast FH/MFSK system differs from a slow FH/MFSK system in that there are multiple hops per M -ary symbol
- Hence, in a fast FH/MFSK system, each hop is a chip
- In general, fast-frequency hopping is used to defeat a smart jammer's tactic that involves two functions:
 - measurements of the spectral content of the transmitted signal,
 - retuning of the interfering signal to that portion of the frequency band
- To overcome the jammer, the transmitted signal must be hopped to a new carrier frequency before the jammer is able to complete the processing of these two functions



Fast Frequency Hopping

- At the receiver, noncoherent detection is used
- However, the detection procedure is quite different from that used in a slow FH/MFSK receiver
- In particular, two procedures may be considered:
 1. For each FH/MFSK symbol, separate decisions are made on the K frequency-hop chips received, and a simple rule based on majority vote is used to make an estimate of the dehopped MFSK symbol
 2. For each FH/MFSK symbol, likelihood functions are computed as functions of the total signal received over K chips, and the largest one is selected



Fast Frequency Hopping

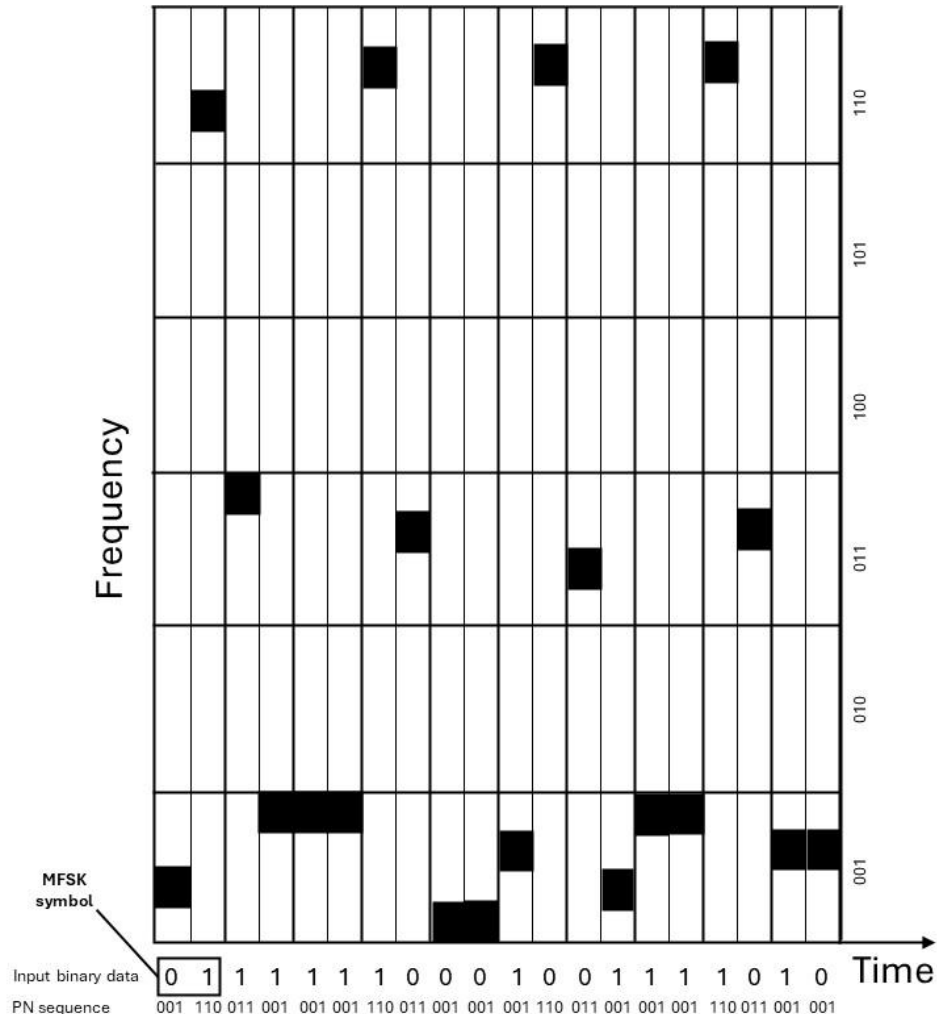
- Example:
- Variation of the transmitted frequency of a fast FH/MFSK signal with time
- The signal has the following parameters:

Number of bits per MFSK symbol	$K = 2$
Number of MFSK tones	$M = 2^K = 4$
Length of PN segment per hop	$k = 3$
Total number of frequency hops	$2^k = 8$



Fast Frequency Hopping

- Example:

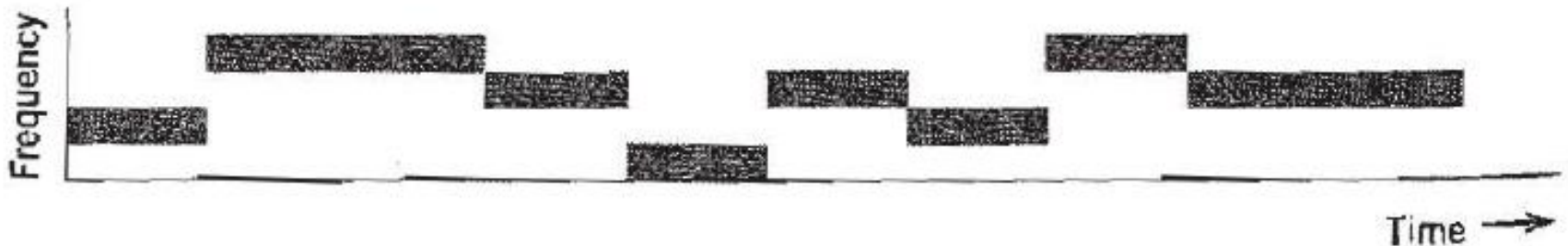


Fast Frequency Hopping

- In this example, each MFSK symbol has the same number of bits and chips, that is, the chip rate R_c is the same as the bit rate R_b
- After each chip, the carrier frequency of the transmitted MFSK signal is hopped to a different value, except for few occasions when the k -chip segment of the PN sequence repeats itself

Fast Frequency Hopping

- Example:
- Next figure depicts the time variation of the frequency of the dehopped MFSK signal, which is the same as that in slide 59



Code Division Multiplexing – CDM

Code Division Multiple Access - CDMA

- Code division multiplexing (CDM) is a multiplexing technique that uses spread spectrum
- In spread spectrum communications, a narrowband signal is spread over a larger band of frequency or across multiple channels
- It does not constrict bandwidth's digital signals or frequencies
- It is less susceptible to interference, thus providing better data communication capability and a more secure private line



Code Division Multiplexing – CDM

Code Division Multiple Access - CDMA

- When CDM is used to allow multiple signals from multiple users to share a common communication channel, the technology is called Code Division Multiple Access (CDMA)
- To each group of users is given a shared code and individual conversations are encoded in a digital sequence
- Data is available on the shared channel, but only those users associated with a particular code can access the data



Code Division Multiple Access - CDMA

- Each communicating station is assigned a unique code.
The codes stations have the following properties:
 - If code of one station is multiplied by the code of another station, it yields 0 – the codes are orthogonal – Inner product is 0
 - If code of one station is multiplied by itself, it yields a positive number equal to the number of stations
- The technique can be explained by the following example:
 - Consider that there are four stations w , x , y and z that have been assigned the codes c_w , c_x , c_y and c_z and need to transmit data d_w , d_x , d_y and d_z respectively



Code Division Multiple Access - CDMA

- The technique can be explained by the following example:
 - Each station multiplies its code with its data and the sum of all the terms is transmitted in the communication channel
 - Thus, the data in the communication channel is $d_w.c_w + d_x.c_x + d_y.c_y + d_z.c_z$
 - Suppose that at the receiving end, station z wants to receive data sent by station y. In order to retrieve the data, it will multiply the received data by the code of station y which is c_y
 - $$\begin{aligned} \text{data} &= (d_w.c_w + d_x.c_x + d_y.c_y + d_z.c_z).c_y = d_w.c_w.c_y + d_x.c_x.c_y + d_y.c_y.c_y + d_z.c_z.c_y \\ &= 0 + 0 + d_y.4 + 0 = 4.d_y \end{aligned}$$
 - Thus, it can be seen that station z has received data from only station y while neglecting the other codes



Code Division Multiple Access - CDMA

- The properties are:
 - If a sequence is multiplied by itself, the result is m , i.e. the number of stations.
 - If a sequence is multiplied by another sequence, the result is 0
 - For adding two sequences, we add the corresponding positional elements
- Let us ascertain the above properties through an example



Code Division Multiple Access - CDMA

- Consider the following chip sequences for the four stations w, x, y and z
- $[+1 -1 -1 +1]$, $[+1 +1 -1 -1]$, $[+1 -1 +1 -1]$ and $[+1 +1 +1 +1]$
- Each sequence has four elements.
- If $[+1 -1 -1 +1]$ is multiplied $[+1 +1 -1 -1]$, we get 0
- If $[+1 -1 -1 +1]$ is multiplied by itself, i.e. $[+1 -1 -1 +1].[+1 -1 -1 +1]$, we get $+1+1+1+1 = 4$, which is equal to the number of stations
 - Commonly used orthogonal codes are Walsh-Hadamard codes



Code Division Multiple Access - CDMA

- Each system can use a different Walsh-Hadamard sequence to spread its data sequence
- This way all users can communicate using the same frequency band and transmit simultaneously
- The other users signal is like noise
- CDM and CDMA are often used in mobile phone communications and GPS



Reading Material

- *Simon Haykin, Communication Systems, McMaster Univ, 5th Edition, 2009 (ISBN 978-047-16979-0-9)*
- *A. Bruce Carlson and Paul Crilly. Communication Systems – An Introduction to Signals and Noise, McGraw-Hill, 5th edition, 2009 (ISBN 978-007-338-040-7)*
- *Bernard Sklar and Fredric Harris, Digital Communications: Fundamentals and Applications, Pearson, 3rd Edition, 2020 (ISBN-13: 978-0134588568)*



Reading Material

